

PROGRAM 1: FACTORIAL OF A NUMBER

```
fact=function(num){  
  if (num==0||num==1){  
    return(1)  
  }else{
```

PROGRAM 2: PALINDROME CHECKER

```
palin<-function(result){  
  og<-result  
  rev<-0  
  while(result>0){  
    a<-result%%10  
    rev<-(rev*10)+a  
  
    if(og==rev){  
      cat("It is a pallindrome ")
```

PROGRAM 3: LINEAR REGRESSION

```
data(mtcars)
model=lm(mpg~wt,data=mtcars)
print(summary(model))

newcars<-data.frame(wt=c(0.2,0.3,0.5))
predicted=predict(model,newdata=newcars)
results=data.frame(weight1000lbs=newcars$wt,) Tj 0 -10 Td (          PREDICTED_MG=predicted)
print(results)

plot(mtcars$wt,mtcars$mpg,) Tj 0 -10 Td (main="MPG VS WEIGHT", xlab="MPG",) Tj 0 -10 Td (ylab="WEIGHT", pch=14, col="red", lwd=2)
```

OUTPUT:

```
Call: lm(formula = mpg ~ wt, data = mtcars)
Coefficients: (Intercept)=37.29, wt=-5.34
R-squared: 0.753
Predictions: weight1000lbs | PREDICTED_MG
1 0.2 | 38.02, 2 0.3 | 36.68, 3 0.5 | 34.00
[Plot: MPG vs WEIGHT with regression line]
```

PROGRAM 4: LOGISTIC REGRESSION

```
data<-read.csv("student_data.csv")
print(data)
model<-glm(Result ~ Hours, data=data,) Tj 0 -9 Td (          family=binomial)
cat("Logistic Regression Model Summary: ")
print(summary(model))
new_data<-data.frame(Hours=5.5)
probability<-predict(model, new_data,) Tj 0 -9 Td (          type="response")
cat(" Predicted Probability of passing:",) Tj 0 -9 Td (    probability," ")
prediction<-ifelse(probability>=0.5,1,0)
cat("Predicted Class(0=Fail,1=Pass):",) Tj 0 -9 Td (    prediction," ")
class<-ifelse(prediction==1,"Pass","Fail")
cat("Final Prediction:",class," ")
```

OUTPUT:

```
Call: glm(formula = Result ~ Hours, family=binomial)
Coefficients: (Intercept)=-4.08, Hours=0.824
Residual deviance: 38.23, AIC: 42.23
Predicted Probability of passing: 0.7342
Predicted Class: 1
Final Prediction: Pass
```

PROGRAM 5: VARIANCE, COVARIANCE & CORRELATION

```
data(mtcars)

# Selecting relevant columns
data_subset <- mtcars[, c("mpg", "wt", "hp")]

cat("Variance: ")
print(var(data_subset))

cat(" Covariance Matrix: ")
print(cov(data_subset))

cat(" Correlation Matrix: ")
print(cor(data_subset))
```

OUTPUT:

```
Variance: mpg=36.32, wt=9.77, hp=4700.87
Covariance: mpg-wt=-5.12, mpg-hp=-320.73, wt-hp=44.19
Correlation: mpg-wt=-0.868, mpg-hp=-0.776, wt-hp=0.659
```